MMALS-CAL: Regime-Aware Conformal Evidence Verification for Continual-Learning Systems

Version 0.3.2 - Offline Cross-Benchmark Evidence Synthesis

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Abstract

Continual-learning evaluations emphasize accuracy, retention, and forgetting, but a deployed system must answer an additional question: is its current prediction supported by evidence that is valid for the present regime and sufficiently precise to justify action? We introduce MMALS-CAL, a standalone companion verification layer within the MMALS program. It separates predictive competence, context recognition, marginal coverage, prediction-set compactness, and operational actionability. We execute a harmonized offline replay over seven final-checkpoint selected-policy traces comprising 250,000 observations, five seeds per benchmark, and five calibration strategies. Fresh oracle-regime calibration reaches aggregate coverage between 0.8889 and 0.9062 for a 0.90 target, while singleton action rates range from 0.15% on CORe50 to 92.43% on SplitCIFAR10. Regime recognition remains a distinct requirement: on CORe50, coverage decreases from 0.9011 under fresh oracle-regime calibration to 0.7255 when the same regime-specific calibrators are selected by inferred context, a 17.6 percentage-point loss. SplitCIFAR10, SplitCIFAR100, and CORe50 use frozen small-CNN feature representations, so these results are conditional on those upstream representations. The singleton gate is a conservative baseline for autonomous single-label actionability, not a universal deployment policy. MMALS-CAL is not formal verification or a proof of universal safety; it is a fail-closed statistical evidence-verification framework that makes assumptions, local failures, abstention, and prohibited claims explicit.

Keywords: continual learning, conformal prediction, calibration, regime shift, context inference, selective prediction, abstention, evidence verification.

1 Introduction

A continual-learning system can retain earlier tasks and still be unsafe to use. It may apply an old confidence rule after a distribution shift, recall a calibrator associated with the wrong context, or obtain nominal coverage only by returning prediction sets too broad to support a useful decision. Accuracy and forgetting do not establish whether the evidence supporting the next action is current, local, and sufficiently precise.

MMALS-CAL turns the statement

> "I have learned, I recognize the context, and I do—or do not—have enough fresh evidence to act."

into four separately testable obligations. The central contribution is the five-axis decomposition: competence, coverage, compactness, context, and ACTION.

2 Origin and Standalone Positioning

The MMALS Chronicle describes a mycelium-inspired continual-learning architecture with inferred context, specialized hosts, routing, and route-function memory. The host organisms correspond to the trees; the fungal system is the mediating mycelial network; the mycorrhizal interface is the exchange relationship between hosts and fungal medium. The analogy motivates distributed specialization but supplies no statistical guarantee.

MMALS-CAL is a companion in provenance but standalone in exposition. It does not train MMALS. It consumes exported probabilities, labels, task/regime identifiers used for offline analysis, and inferred-context evidence, then asks whether the uncertainty rule associated with the active situation remains valid and actionable.

3 Position in the Verification Framework

The wider evidence framework separates:

Metric Evaluator Computes accuracy, forgetting, coverage, set size, context quality, and action rate.

Protocol Verifier Checks provenance, no leakage, deterministic selection, and comparable units.

MMALS-CAL Tests regime-aware calibration, compactness, and ACTION/NO-DECISION behavior.

Claim Verifier Maps conclusions to supported, unsupported, or prohibited statements.

The first, second, and fourth layers verify the scientific work. MMALS-CAL additionally prepares the logic required by a future runtime evidence controller. Version 0.3.2 remains an offline final-checkpoint replay.

4 Component Status

Component	v0.3.2 status	Evidence and remaining gap
A. CUSUM/BOCPD regime detection	Absent online	Regime labels and exported context are replayed; no autonomous stream detector.
B. No-leakage buffer	Implemented offline	Deterministic 25/75 reserved-calibration/deployment split independent of scores.
C. Regime-aware conformal quantile	Implemented offline	Five strategies, including stale, oracle, inferred-bucket, and context-selected recall.
D. ACTION/NO-DECISION gate	Operational but simplified	Singleton gate is a conservative common baseline, not a final deployment policy.

5 Method

5.1 The conformal calibration quantile

A quantile is a threshold below which a chosen fraction of sorted values falls. MMALS-CAL sorts calibration nonconformity scores

$$s(x, y) = 1 - p(y \mid x). \tag{1}$$

For n calibration scores, error level α, and sorted scores $s_{(1)} \leq \cdots \leq s_{(n)}$, the finite-sample corrected index and quantile are

$$k = \min\left(n, \lceil (n+1)(1-\alpha) \rceil\right), \qquad \hat{q} = s_{(k)}. \tag{2}$$

The prediction set is

$$\Gamma(x) = \{\, y : 1 - p(y \mid x) \leq \hat{q} \,\}. \tag{3}$$

A larger $\hat{q}$ usually improves coverage by admitting more labels, but can reduce compactness and actionability. The finite-sample marginal guarantee relies on exchangeability between calibration and deployment observations; it is not automatically a conditional guarantee for every task, class, seed, or time interval.

5.2 No-leakage offline protocol

Each seed-task-class group is partitioned deterministically into 25% reserved calibration and 75% deployment. Selection is independent of probabilities, predictions, correctness, and context confidence. The release contains all seven compressed filtered traces and the complete replay implementation.

5.3 Five calibration strategies

We compare one global calibrator per seed; a T0 quantile reused as stale evidence; fresh oracle-regime calibration; calibration in inferred-context buckets; and oracle regime quantiles selected by inferred context. The last strategy isolates the damage caused by recalling a valid calibrator under the wrong regime.

5.4 Action gate

The common research rule is

$$\mathrm{decision}(x) = \begin{cases} \texttt{ACTION}, & |\Gamma(x)| = 1, \\ \texttt{NO\_DECISION}, & \text{otherwise.} \end{cases} \tag{4}$$

This is a conservative baseline for autonomous single-label actionability. It is not the maximum achievable action rate, a universal product policy, or a formal lower bound without specifying the admissible policy family and its costs.

6 Semantic Foundations and Research Dialogue

This section preserves the full translated research dialogue because the gradual clarification of the terms is itself part of the methodology. The purpose is not only to add definitions, but to make explicit how those definitions change the way the MMALS-CAL problem is understood, defended, and extended.

6.1 The conformal calibration quantile - definition

Starting point: what is a quantile in the general sense?

Before discussing conformal calibration, let us recall what a quantile is. If we take a list of numbers and sort them from smallest to largest, the 90% quantile is the value below which 90% of the numbers in the list fall. For example, if we sort 100 scores and the 90% quantile is 0.42, this means that 90 of those 100 scores are less than or equal to 0.42, and 10 are greater than it.

The score being sorted here: the nonconformity score

In MMALS-CAL, for each observation `x` whose true class `y` is known, we compute a nonconformity score:

$$s(x, y) = 1 - p(y \mid x)$$

where `p(y | x)` is the probability assigned by the model to the true class `y` for that observation. Concretely: if the model was highly confident and correct, `p(y|x)` is close to 1, so the score `s` is close to 0 - little deviation, little "nonconformity." If the model was surprised by the true answer, meaning that it assigned a low probability to the correct class, the score `s` is close to 1 - a large deviation.

This score is computed on every observation in the calibration set - that is, the dataset reserved exclusively for this purpose, never used by the model during training, and distinct from the deployment set (see component B on the no-leakage buffer).

The conformal quantile itself

All of these nonconformity scores are sorted from smallest to largest. For a target coverage level - say 90%, which corresponds to a tolerated error level $\alpha = 0.10$ - we look for the value located just above 90% of the observed scores. Formally, for `n` calibration observations:

$$k = \min\left(n, \lceil (n+1)(1-\alpha) \rceil\right)$$

$$\hat{q} = s_{(k)}$$

In other words, we take the k-th smallest score in the sorted list, with a small adjustment - `(n+1)` rather than `n` - that guarantees the statistical property even for small samples. This value $\hat{q}$ is the **conformal quantile**.

How this quantile becomes a decision threshold

Once $\hat{q}$ has been computed, it is used to construct the prediction set for a new observation: the set includes **all** classes whose nonconformity score would be less than or equal to $\hat{q}$:

$$\Gamma(x) = \{y : 1 - p(y \mid x) \leq \hat{q}\}$$

In other words, $\hat{q}$ sets the tolerance bar. The smaller $\hat{q}$ is, the more restrictive the prediction set will be: few classes are included, potentially only one singleton, so ACTION may be possible. The larger $\hat{q}$ is, the broader the set will be: several classes are included, so NO-DECISION becomes more likely, but the coverage guarantee becomes more robust.

Why "conformal," and why the guarantee holds

The term comes from *conformal prediction* (Vovk et al., 2005): a new observation is said to be "conformal" with the rest of the set when its nonconformity score does not deviate too much from the scores already observed during calibration. The statistical guarantee - that the true class belongs to the set $\Gamma(x)$ in at least 90% of cases - relies on an assumption called *exchangeability*: the calibration and deployment observations must come from the same distribution, that is, from the same regime.

Why "per regime" changes everything

This is exactly where the preceding topic - the regime-aware conformal quantile - connects to this one. If $\hat{q}$ is computed on data that mix several different regimes, the exchangeability assumption is locally violated: the global quantile may be valid on average, but too optimistic for a difficult regime and too conservative for an easy regime. Recomputing a distinct $\hat{q}$ for each regime - using only the calibration observations assigned to that regime -

restores the validity of the guarantee at the local level, which is precisely what MMALS-CAL tests through its five calibration strategies.

6.2 Is regime-change detection therefore fundamental in continual learning?

> Regime-change detection is therefore fundamental in continual learning, especially because I imagine that industrial or production continual-learning systems learn and infer continuously, update themselves continuously, partly on known regimes that have changed, or on genuinely new regimes with no overlap - or only partial overlap - with what already exists. This is what we would like to achieve: not restart from zero for every learning episode, but progressively reuse whatever overlaps with existing knowledge.

Yes - and I think this identifies something structurally important that goes beyond MMALS-CAL itself: regime-change detection is not one feature among others in continual learning; it is the condition that makes almost everything else valid.

Why it is fundamental rather than optional

In continual learning, the system learns under an implicit assumption of local stability - "what I learned remains valid as long as nothing has changed." Yet this assumption is rarely tested explicitly in most conventional continual-learning architectures. They focus on retention, meaning not forgetting, and transfer, meaning reusing prior knowledge, without ever asking: "Am I still in a context where my current confidence rule applies?"

The result is exactly what CAL demonstrates empirically: a valid calibrator applied under the wrong regime is not safe. This is not a calibration defect; it is a *detection* defect. Calibration can repair only what detection has correctly identified upstream.

The industrial case described here is precisely the difficult case

Two very different situations arise in a production system.

The first case - a return to a known regime - is theoretically the simplest, but it is also where systems often fail silently. The paper demonstrates this with CORe50: earlier contexts are frequently misrecognized, so the system believes it is in a new regime, or worse, believes it is in the wrong known regime and applies an invalid quantile without producing any obvious alarm. The displayed confidence remains plausible. This is the most dangerous scenario because it does not generate an obvious error signal; it generates false confidence.

The second case - a new regime that partially overlaps with existing ones - is the objective being formulated here: do not restart from zero, but also do not pretend that the old calibrator applies unchanged. This is exactly the problem that CUSUM and BOCPD address in different ways.

CUSUM is binary: it declares a change or no change. It says nothing about the degree of overlap with the previous regime.

BOCPD is more interesting for this precise case because it maintains a probability distribution over the *run length*. It can therefore express graded uncertainty - "probably a new regime, but with a 30% chance that this is still a variant of the previous one" - rather than forcing an immediate hard decision. This is closer to what partial overlap requires: the system does not need to decide immediately between "old" and "new"; it can remain in a `SUSPECTED_SHIFT` state while calibration is progressively reconstructed.

The deeper problem this reveals

What is being described - learning continuously without restarting from zero, while partially overlapping with existing knowledge - requires a notion of regime that is not binary, known versus unknown, but *metric*: how far am I from the closest regime that I already know?

This is where Geometry-MMALS becomes relevant, as the status document already noted: distance in latent space, distance between route distributions, local curvature. Change detection through CUSUM or BOCPD answers "Has something changed?" but does not answer "Does this new regime resemble an already calibrated regime, and by how much?"

Concretely, this suggests that component A as described in the paper - binary states such as stable, suspected, and calibrating - is probably an intermediate stage. The actual industrial target is a detector that does not produce only a flag, but a distance or a probability of overlap with every known regime. That would allow the system to choose among three actions rather than two: reuse an existing calibrator, recalibrate from a nearby calibrator as a warm start, or calibrate entirely from scratch.

> Is this the type of graded overlap mechanism envisioned for the roadmap after stabilizing CAL v0.3.1?

> Yes, Claude, this is exactly where I want to go. I want to understand how geometric proximity coupled with a probability of change can be used, how the system should react according to distance and/or change probability, how it can identify which known regime or regimes the new situation might correspond to, and therefore which routes and functions - including hosts - should be reused, retrained, or readapted.

This is a precise formulation of the problem, and it has a three-stage structure that is worth separating clearly before considering how the stages interact, because they are three mathematically different objects that are easily confused.

6.2.1 Stage 1 - The change signal: the probability of change

This is what CUSUM or BOCPD produces: has something changed, and with what confidence? It is a scalar signal or a distribution over the run length. It says nothing about the *nature* of the change, only about its existence.

6.2.2 Stage 2 - Geometric proximity: where are we, relative to what?

This is what Geometry-MMALS can contribute. Once a change is suspected, where is the new observation located in relation to the space of previously known regimes? This is a distance - in latent space, route space, or the functional space of hosts. It is no longer binary; it is metric: close to regime 3, halfway between regimes 3 and 7, or far from every known regime.

6.2.3 Stage 3 - The reuse decision: what action should be taken on which structure?

This is the new part being introduced, and it is the part that is most absent today: how should the combination of the first two stages determine what to do with existing routes and hosts?

6.3 How the change signal and geometric distance should combine

The key point is that the two signals are not interchangeable. They answer orthogonal questions. A strong change signal with a short geometric distance is different from a strong change signal with a large distance, and both are different from a weak change signal with a short distance. The problem therefore forms a two-axis map rather than one scalar score.

When the change signal is weak - the system is probably still in the current regime - and the distance to the current regime is small, nothing needs to be done. This is the stable case; the current quantile remains valid.

When the change signal is strong but the distance to the closest known regime is small, this is exactly the partial-overlap case. The system has changed, but not toward the unknown: it has moved toward something resembling what it already knows, or toward a combination of known regimes. This is the most interesting industrial case because it allows a graded reaction rather than a binary choice.

When the change signal is strong and the distance to every known regime is large, the situation is a genuinely new regime. Full recalibration is required, and potentially a new host must be trained.

When the change signal is weak but the distance increases progressively - slow drift without a sharp breakpoint - this is a case that neither CUSUM nor BOCPD captures well on its own. It is instead a continuous-drift signal that should trigger periodic revalidation of the quantile even when no breakpoint has been declared.

6.4 Reusing routes and hosts: three reaction regimes, not two

The implicit insight is that between "reuse everything unchanged" and "retrain everything from scratch," there is an intermediate level that is probably the real added value of the approach.

Direct reuse - The distance to one known regime is small. Recalibrate that regime's existing quantile without changing the route or the host. This is the trivial case already covered by the paper's `context-selected oracle` strategy, provided that context recognition is reliable.

Recombination or warm start - The distance is small to several known regimes simultaneously, or intermediate to one regime. In this case, the existing route probably should not be discarded, but the quantile must be reconstructed from a mixture or interpolation of the calibrators of nearby regimes rather than rebuilt entirely from scratch. This is the conformal equivalent of fine-tuning rather than new training. The open question is how to weight the contribution of each nearby regime: by geometric distance itself, by a combination of distance and change confidence, or by another learned or calibrated principle?

New formation - The distance to every known regime is large. A new host or route is required, along with entirely fresh calibration. This case is already covered by CAL under `fresh oracle calibration`, but without the geometric trigger that would decide when to use it rather than attempting a recombination.

6.5 What is most concretely missing to formalize this

The hardest point is probably not change detection or geometric distance separately; both already exist as components - CUSUM or BOCPD on one side, Geometry-MMALS on the other. The difficult part is the decision function that transforms `(change probability, distance to known regimes)` into one of the three actions above, with a threshold that is itself statistically calibrated. Otherwise, the system falls back into the same problem as the ACTION/NO-DECISION gate: an unverified heuristic rule.

One possible direction is to formulate this as a multiclass classification problem over the space `(change score , distance)`, where the classes are exactly the three reaction regimes. That decision could itself be conformally calibrated, in the same spirit as CAL but applied one level higher, to the routing or reuse decision rather than to the final classification decision.

> Should this be seen as an extension of CAL - CAL applied to the reuse decision - or as a separate component positioned before CAL in the pipeline?
>
> We must first distinguish offline CAL from online CAL. I think offline CAL will always remain useful and should continue to be maintained and documented. We will probably need to document, and perhaps standardize, the parameters and signals that should be recorded during learning, inference, or decision runs for every possible component in a continual-learning, reinforcement-learning, or broader adaptive system. This would make the tool generic, without constantly having to return to `.pt` files for PyTorch, other checkpoint formats for Keras, and so on. At what level in the online pipeline would the multiclass classifier over this space be placed? And, following the same "keep it simple" principle, do we really need one more component? Could there instead be a natural mathematical inference and routing mechanism without adding another rule box that itself needs calibration, even if statistically?

This question is excellent and touches something fundamental about the architecture.

6.6 First, the offline-versus-online distinction

Offline CAL has permanent and independent value. It answers a different question from online CAL.

Offline CAL asks: *Is the calibration structure that I designed coherent, uncontaminated, and capable of producing measurable guarantees on historical traces?* It is an audit tool, an experimental-validation tool, and a scientific-publication tool. It does not disappear when an online loop is added; instead, it becomes the test bench used to validate whether the online version behaves as expected.

Online CAL asks: *At the moment when I observe this datum, is my confidence rule still valid, and what decision should I take now?* It is a runtime component.

The two must coexist exactly as in other engineering domains: the test bench does not disappear when the engine runs in production.

6.7 Standardizing signals may be the most durable contribution

The point being raised implicitly is a real and under-addressed problem in continual-learning research: no standard format exists for what a learning, inference, or decision run must export to be independently auditable, regardless of framework.

Concretely, each component in the pipeline - whether host, router, memory, or gate - should export a minimal set of signals in a neutral format, independent of PyTorch, Keras, JAX, or any other framework. Something like:

```
observation_id
timestamp
regime_id_inferred
regime_confidence
nonconformity_score
prediction_set
action_declared
route_used
host_used
calibrator_id
calibrator_age
```

This neutral format would allow offline CAL to apply to any continual-learning system, not only MMALS. It is a generic contribution with real standardization value. It is also "keep it simple" in the best sense: no dependency on `.pt` checkpoints or internal framework-specific structures, only clean signals emitted by each component.

This should probably become a dedicated section in the CAL documentation, or even a separate specification document.

6.8 Where the multiclass reuse classifier would sit in the online pipeline

The question is precise: where should it be placed, and is another explicit box really necessary?

In the current pipeline:

```
Incoming observation
    |
RC2O - context and route inference
    |
Selected host - prediction
    |
CAL - conformal quantile + ACTION/NO-DECISION gate
```

The open question is where the decision "reuse / recombine / retrain" belongs. The deeper question is whether that decision requires an explicit additional component, or whether it should emerge naturally from RC2O routing coupled with geometry.

The honest answer is: it depends on what RC2O already produces.

If RC2O produces not only one inferred regime but a *confidence distribution over several regimes simultaneously* - for example, "60% regime 3, 30% regime 7, 10% unknown" - then the recombination decision does not require another component. It is already implicit in the distribution: short distance to one dominant regime implies direct reuse; probability mass spread across several nearby regimes implies recombination; probability mass on `unknown` implies new formation. Natural routing *is* the decision.

If RC2O produces only one regime and a scalar confidence, the information required for recombination is missing. RC2O must then be enriched, or an explicit layer must be added.

Therefore the real question is not "Do we need one more box?" but "Does RC2O already produce a distribution over regimes, or only an argmax?"

6.9 Natural mathematical routing without an additional rule box

This is the deepest question, and it has a partially honest answer.

Conformal prediction itself is a form of natural routing in a precise sense: it does not choose one class, but includes every class compatible with the available evidence. It already lets the mathematics determine the degree of certainty without a heuristic rule.

The problem is that this property does not yet exist at the *meta-routing* level - the decision over regimes and calibrators. At present, this level still uses rules such as thresholds τ on regime confidence, freshness criteria, and coverage tests. These rules are reasonable, but heuristic.

The direction toward "fewer boxes, more mathematics" would be to apply conformal prediction one level higher: produce not one calibrator, but a *set of calibrators compatible with the current geometric evidence*, weighted by their distance to the observed regime. The final quantile would be an aggregation of those calibrators, which would amount to an automatic warm start without a binary rule deciding which calibrator to choose.

This remains to be formalized, but the principle is consistent with "keep it simple" in the sense that it does not add another rule box; it extends the same conformal mechanism to a higher level of abstraction.

> Should we begin by sketching the standardized signal format - the interface contract that every pipeline component should satisfy in order to make CAL generic?
>
> Before doing that, we should first review everything MMALS already generates as signals. There is a profusion of them: EER-like metrics, LSA, Hydro, and many others. We should analyze an existing run and the different playbooks and results: MMALS core with RC2O, MMALS Geometry, mmals-tput, the MMALS Goal-Adaptive Geo/RL/Forward-Backward package, mmals-v1.1-continual-learning-evidence - which should be similar to MMALS core but actually adds the continual-learning layer and analyzes losses caused by learning other regimes over time - and synthetic-data-scaling-mmals, a public multilevel analysis of synthetic-data scaling and model collapse, inspired by Julia Kempe's lecture, with reproducible experiments and research directions for MMALS, STRAT-Q, probabilistic MTP, and related work.
>
> Additional repositories include:
>
> - synthetic-data-scaling-mmals - **Public.** Multilevel analysis of synthetic-data scaling and model collapse, inspired by Julia Kempe's lecture, with reproducible experiments and research directions for MMALS, STRAT-Q, probabilistic MTP, and related work. **Primary repository language:** TeX. **License shown:** Other. **Status at the time of the dialogue:** updated yesterday.
> - geometry-mmalls-g1 - **Public.** Grounded functional geometry for MMALS continual learning: context manifolds, route-simplex dynamics, causal host specialization, and functional memory transport. **Primary repository language:** Jupyter Notebook. **License shown:** MIT License. **Status at the time of the dialogue:** updated yesterday.
> - mmals-stable-plastic-context-memory - **Public.** Dataset-agnostic and backward-compatible stable-plastic context memory for MMALS, with legacy reproduction, adaptive fusion, deployed-posterior distillation, cross-dataset regression gates, and revision evidence. **Primary repository language:** Jupyter Notebook. **License shown:** Other. **Status at the time of the dialogue:** updated 3 days earlier.
> - mmals-tput - **Public.** Research specification and reference implementation of MMALS-TPUT, unifying latent geometry, energy-guided routing, dual memory, goal-conditioned control, RC2O-to-RL extensions, and Geometry-G1 validation. **Primary repository language:** Python. **License shown:** MIT License. **Status at the time of the dialogue:** updated 3 days earlier.
> - mmals-v1.1-continual-learning-evidence - **Public.** Full MMALS v1.1 research status, consolidating MNIST-family foundations, SplitCIFAR10 robust qualification, SplitCIFAR100 and CORe50 diagnostics, and the CORe50 v2.3 rescue ablation. **Primary repository language:** Jupyter Notebook. **License shown:** Other. **Status at the time of the dialogue:** updated 4 days earlier.
> - mmals-goal-adaptive-control - **Public.** Goal-adaptive Geo/RL/Forward-Backward control layer for MMALS, extending RC2O routing with objective-conditioned route selection under accuracy, retention, cost, drift, and specialization trade-offs. **Primary repository language:** Python. **License shown:** Other. **Status at the time of the dialogue:** updated 5 days earlier.
> - mmals-rc2o-splitcifar10-continual-learning - **Public.** External continual-learning bridge experiments for MMALS/RC2O, starting with robust SplitCIFAR10 evidence over frozen A1 visual features and audited policy selection. **Primary repository language:** Jupyter Notebook. **License shown:** Other. **Status at the time of the dialogue:** updated 5 days earlier.
> - mmals-rc2o-prep-selector-regret-regime-audit - **Public.** MMALS RC2O-prep audit: selector regret, feature-only regime estimation, and regime-certified safe deployment for auditable continual learning. **Primary repository language:** Python. **License shown:** Other. **Status at the time of the dialogue:** updated 2 weeks earlier.

That is correct: before designing an ideal interface contract, we must inventory what already exists in the

MMALS ecosystem. Otherwise, we risk reinventing signals that are already available, or worse, proposing a format that conflicts with signals already produced.

6.10 A concrete signal-inventory process

The repositories should be reviewed in a logical order from the most upstream to the most downstream component in the pipeline, in order to map the signals already exported:

```
mmals-v1.1-continual-learning-evidence    <- base CL layer, learning signals
    |
mmals-rc2o-splitcifar10                   <- RC2O, routing and context signals
    |
mmals-rc2o-prep-selector-regret           <- regret, regime, selector audit
    |
geometry-mmalls-g1                        <- geometric signals: distance, curvature
    |
mmals-stable-plastic-context-memory       <- memory, fusion, distillation
    |
mmals-tput                                <- decision, cost, control
    |
mmals-goal-adaptive-control               <- goals, forward-backward control
    |
synthetic-data-scaling-mmals              <- collapse and scaling signals
```

For each repository, the questions to answer are simple:

- Which CSV, JSON, or Parquet files are exported at the end of a run?
- Which columns contain signals relevant to CAL, such as scores, contexts, routes, or decisions?
- Which signals are currently implicit inside `.pt` files or notebook code, but are not yet exported cleanly?

What can be done immediately

An example output file from a recent run should be examined - for example, from `mmals-v1.1-continual-learning-evidence` or `mmals-rc2o-prep-selector-regret`. It could be a result CSV or simply a list of column names. This would allow the mapping to begin from concrete evidence rather than from abstraction.

Alternatively, the inventory can begin from whichever repository is considered the most central or representative of the current pipeline.

> Yes, we will look at that tomorrow. It is late. We should also review the CAL v0.3.2 freeze again to ensure that it properly includes the definitions and semantics that, through these small human-like reflections, help us understand what we are doing and what we would ideally like to achieve. Without that, the project can always collapse into mathematical or metaphysical jargon with no meaning, justified by Colab notebooks, GPU runs, and attractive results that we do not understand and that I cannot defend.
>
> The "end" or milestone of MMALS is also still open. The core has not yet been published - I believe we have been using another, perhaps more scientific term - because I no longer have Colab GPU capacity to finish the CORe50, Split-TinyImageNet, and/or another identified run.

Good night - and yes, this is exactly the right moment to stop.

What has just been formulated is in fact the guiding principle of the entire program:

> **Without defensible semantics, results are not publishable - they are merely attractive.**

This is precisely what distinguishes the intended MMALS approach from many continual-learning works: the goal is to be able to say, "I know what I am measuring, I know why I am measuring it, and I can defend every term in front of a reviewer or a practitioner." CAL v0.3.1 is already a good example of this; the five conclusions in Section 9 are defensible without additional GPU runs.

For the next session - two things should be done in parallel

First, perform the CAL v0.3.2 pass to integrate the definitions and semantics developed here: regime, nonconformity score, conformal quantile, no-leakage buffer, and gate. These are exactly the terms that were missing for the paper to be readable by a non-specialist without becoming condescending to a reviewer.

Second, inventory the signals exported by the existing repositories in order to understand what is truly available before designing anything new.

On the "core" milestone and missing GPU resources

The absence of a complete CORe50 run and Split-TinyImageNet does not necessarily prevent an honest submission, provided the gaps are formulated as explicit scientific limitations rather than concealed omissions. CAL v0.3.1 already does this well for the frozen backbone. The Chronicle can do the same.

> "Chronicle" - where does this term come from? Thank you for the reminder.

The term comes from earlier discussions. It was the name given to **Paper 1**, the first article in the MMALS publication sequence.

The idea behind the word was semantic: a *chronicle* is a faithful and temporally ordered record of what happened - not a theory, not a proof of superiority, but an honest account of a research journey, including controlled negative results and acknowledged limitations.

It was a way to position Paper 1 as:

> "Here is what MMALS is, what we measured, in what order, and what the evidence proves or does not prove."

This contrasts with a paper that would claim to have solved continual learning.

The term also had a practical role in the publication sequence: the Chronicle would establish the arXiv identifier cited by all subsequent papers, including Axis A, Goal-Adaptive, and CAL. It would be the birth certificate of the MMALS research program in the literature.

The name can still be reviewed before submission to determine whether it remains appropriate or whether a more conventional descriptive arXiv title would be preferable. A factual descriptive title may sometimes be easier to position than a programmatic term.

Good night, Quiero.

7 Experimental Scope

The synthesis covers MNIST Robust, FashionMNIST Robust, PermutedMNIST Robust, RotatedMNIST Robust, SplitCIFAR10, SplitCIFAR100, and CORe50. It contains 250,000 selected-policy observations: 62,500 reserved for calibration and 187,500 for deployment. Five seeds and five calibration strategies are evaluated per benchmark.

SplitCIFAR10, SplitCIFAR100, and CORe50 use frozen small-CNN feature representations inherited from the source continual-learning runs. The results therefore characterize the combined upstream representation, learner, and calibration layer; they do not establish backbone invariance or end-to-end raw-pixel continual learning.

8 Expected Behavior of an Evidence-Verification Tool

An effective verifier should not make every input actionable. It should reveal at least four distinct outcomes:

1. a competent model with fresh, compact evidence and frequent action;
2. a competent model with stale evidence and undercoverage;
3. nominal coverage obtained through broad, non-actionable sets;
4. a valid calibrator invalidated by wrong regime recall.

It should also fail closed: absent or insufficient evidence should produce `NO_DECISION` and a recorded reason rather than an unsupported certainty claim.

Table 1: Seven-benchmark profile under fresh oracle-regime calibration. Worst is minimum seed-task coverage.

Benchmark	Top-1	Context	Coverage	Worst	Mean set	ACTION	Acc. if ACTION
MNIST	0.9642	1.0000	0.8889	0.8283	0.90	89.95%	98.82%
FashionMNIST	0.9010	0.9200	0.9001	0.8433	1.03	89.83%	93.58%
PermutedMNIST	0.9698	1.0000	0.9029	0.8617	0.92	91.69%	98.41%
RotatedMNIST	0.9678	1.0000	0.9007	0.8217	0.91	91.32%	98.64%
SplitCIFAR10	0.9140	1.0000	0.9045	0.8617	1.00	92.43%	94.01%
SplitCIFAR100	0.5356	0.9329	0.9062	0.8533	8.53	9.44%	96.09%
CORe50	0.3149	0.7105	0.9011	0.8733	28.67	0.15%	63.66%

9 Cross-Benchmark Results

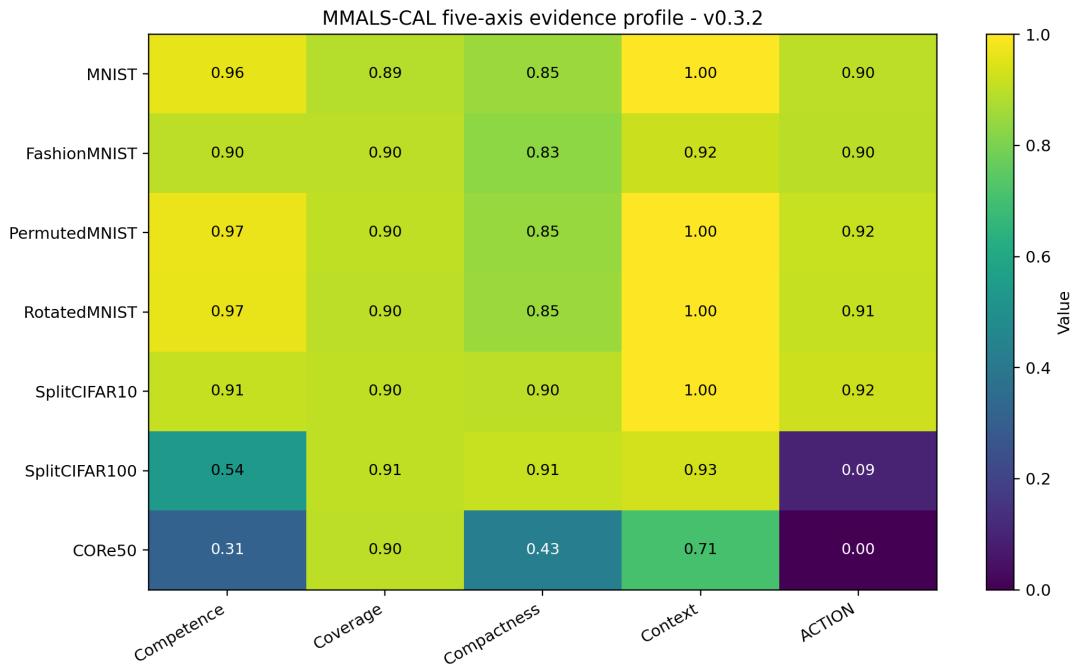

Figure 1: Five-axis profile: competence, coverage, compactness, context, and ACTION.

Fresh oracle-regime calibration approaches the aggregate 0.90 target across all seven benchmarks. The operational meaning differs sharply. SplitCIFAR100 reaches 0.9062 coverage using an average set of 8.53 labels and acts on 9.44% of observations. CORe50 reaches 0.9011 coverage using an average set of 28.67 labels out of 50 and acts on only 0.15%.

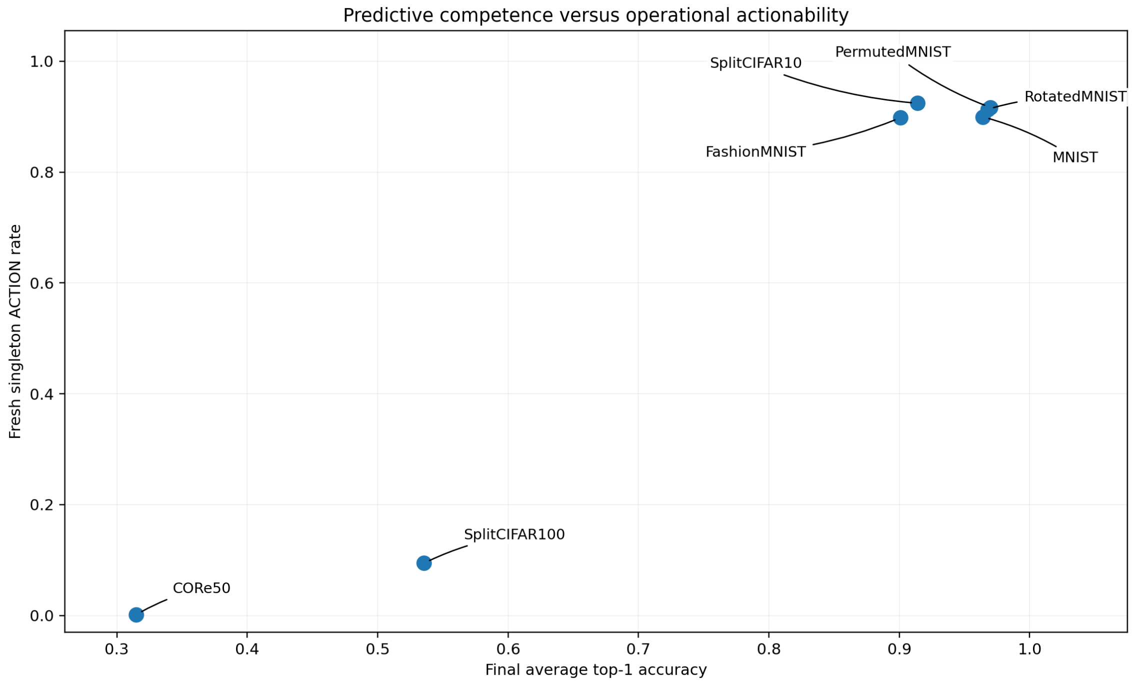

Figure 2: Competence and singleton actionability are related but not interchangeable.

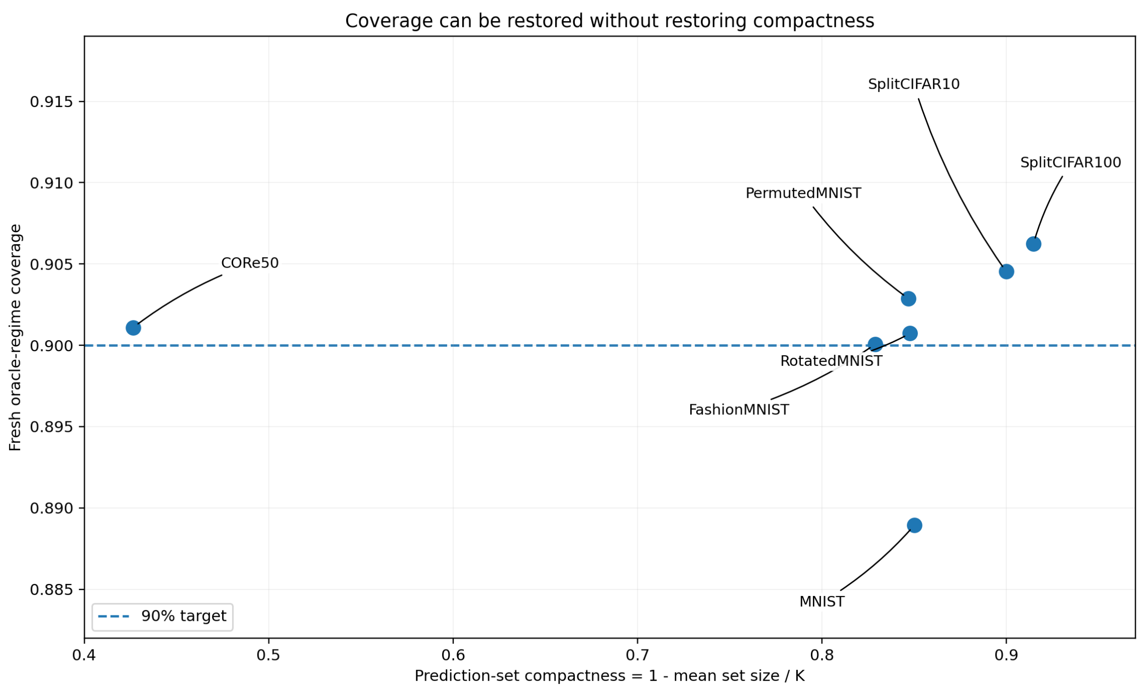

Figure 3: Coverage may be restored without restoring compactness.

9.1 Global averages hide local failures

Aggregate coverage can appear acceptable while individual seed-regime cells under-cover severely. Worst-cell reporting is therefore part of the evidence contract, not an optional diagnostic.

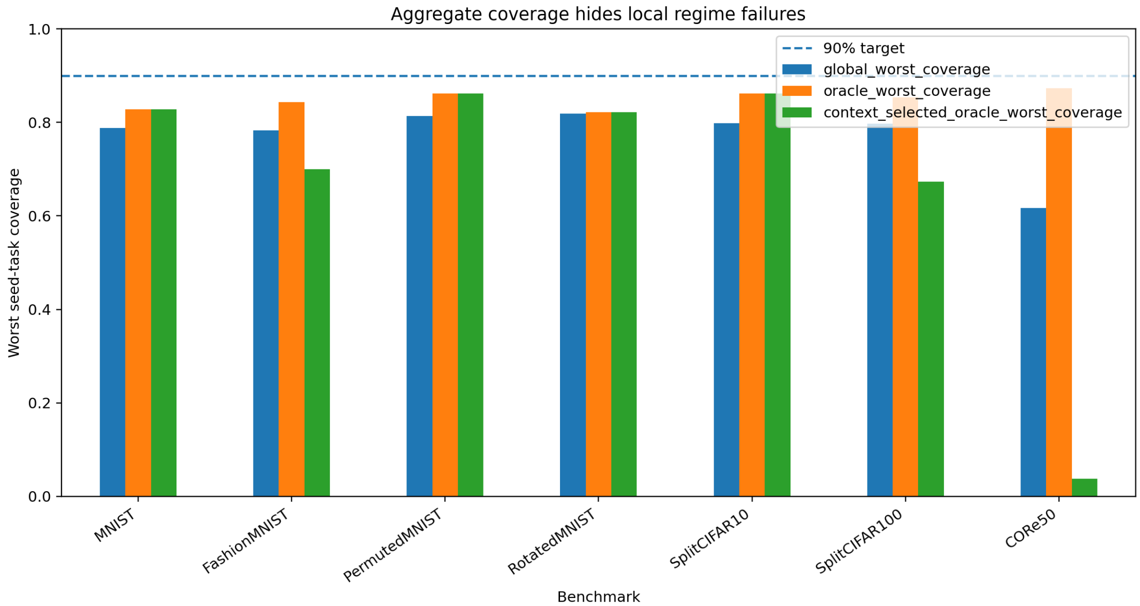

Figure 4: Worst seed-task coverage under global, fresh oracle, and context-selected oracle calibration.

9.2 Stale evidence has two failure modes

Reusing the T0 quantile causes undercoverage on several MNIST-family runs and SplitCIFAR10. On SplitCIFAR100 and CORe50 it becomes highly conservative, retaining broad sets and little actionability. Staleness is a mismatch, not simply overconfidence.

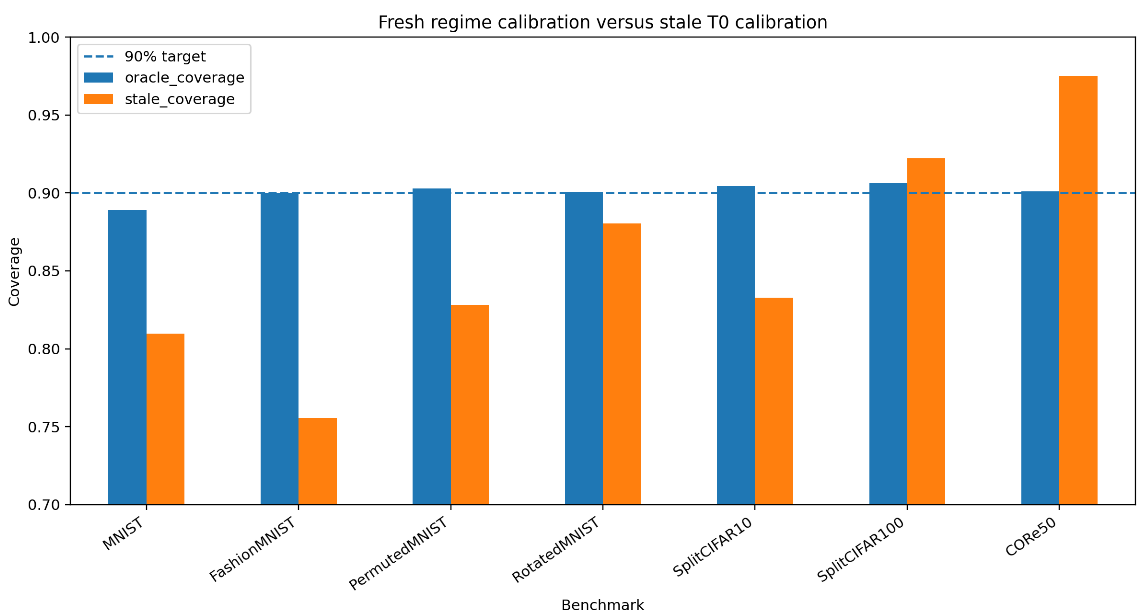

Figure 5: Fresh regime calibration versus stale T0 calibration.

9.3 Wrong context recalls wrong evidence

On CORe50, coverage decreases from 0.9011 under fresh oracle-regime calibration to 0.7255 when the same regime-specific calibrators are selected using inferred context, a 17.6 percentage-point loss. A statistically valid calibrator is not safe when recalled for the wrong situation.

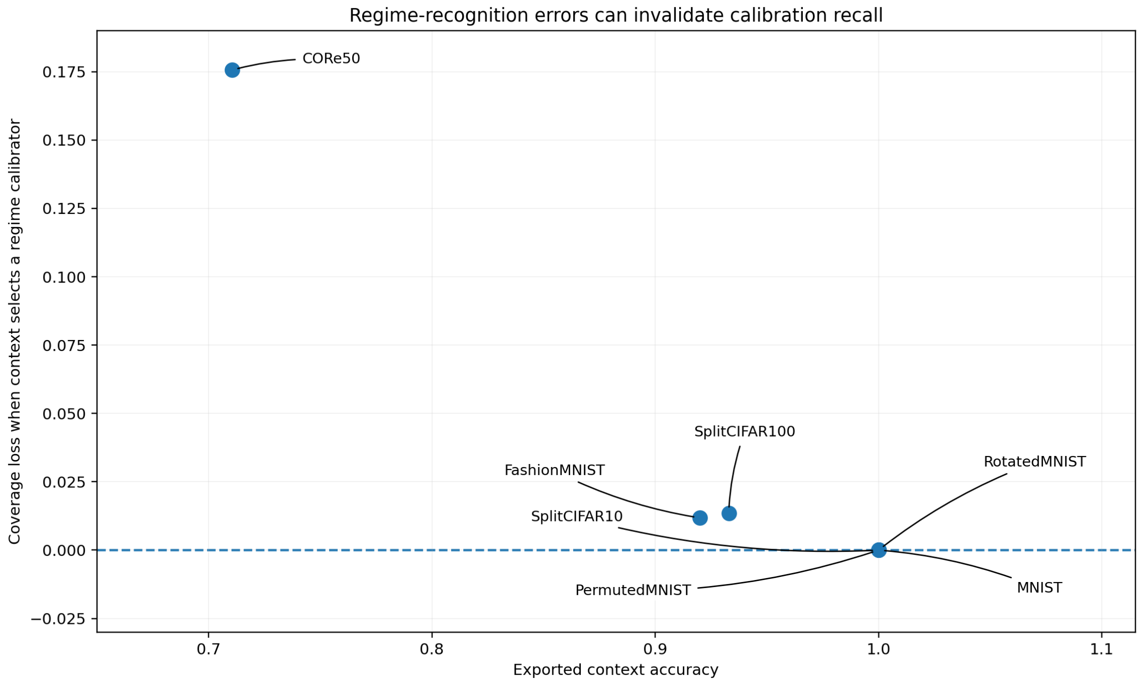

Figure 6: Context errors translate into calibration-recall loss.

10 FashionMNIST RC2O Consistency Patch

The original v0.3 synthesis used an older RC2M-alpha trace. Version 0.3.1 introduced the RC2O-v1 trace; version 0.3.2 preserves that correction in a fully reproducible release. Final average accuracy rises from 0.8872 to 0.9010, context accuracy from 0.88 to 0.92, mean set size falls from 1.0906 to 1.0269, and singleton action rises from 86.48% to 89.83%, while fresh coverage remains approximately 0.90.

> **RC2O does not primarily win on coverage—coverage is imposed by calibration—but on predictive competence, context recognition, prediction-set compactness, and action rate.**

11 What the Results Say About MMALS-CAL

The evidence supports five conclusions: fresh calibration repairs coverage rather than competence; valid calibration can be invalidated by wrong regime recall; global coverage can hide local undercoverage; stale evidence can under-cover or become non-actionably conservative; and actionability is separate from coverage.

These conclusions also define what MMALS-CAL is: an empirical and statistical evidence-verification layer that exposes disagreement between learning, regime recognition, uncertainty, and action. It is not a learned replacement for the backbone and not a universal safety proof.

12 Engineering After the PhD

A production implementation should remain a fail-closed evidence shell around MMALS. It requires a label-free early-warning detector, delayed-label confirmation, an immutable calibration registry, strict temporal separation of detection/calibration/validation windows, a conformal set builder, a cost-aware gate, and an evidence ledger. The online chain should be

$$\text{detect} \rightarrow \text{close} \rightarrow \text{buffer without leakage} \rightarrow \text{recalibrate} \rightarrow \text{validate} \rightarrow \text{reopen}. \tag{5}$$

CUSUM should be an interpretable baseline and BOCPD a probabilistic comparator. TPUT/STRAT-Q may later optimize wrong-action, abstention, evidence, latency, and retention costs while keeping the conformal constraint explicit.

13 For a 12-Year-Old

Imagine a puzzle team with different specialists. The MMALS mycelial network helps send each puzzle to useful specialists. MMALS-CAL is the referee. It asks whether the team learned the puzzle, recognized its type, still has a fresh confidence rule, and has one clear answer. The conformal quantile is a line drawn across earlier "surprise scores." Answers that are not more surprising than the line enter a candidate set. One candidate can mean ACTION; several candidates mean "I am not sure yet." Saying "one of 29 labels" can be honest but not useful enough to act.

14 Limitations

This release is an offline replay of exported final-checkpoint traces. It does not implement online label-free CUSUM or BOCPD, delayed labels, streaming role assignment, live calibration expiry, or automatic gate reopening. Context values are exported by source runs and must still be audited for independence from explicit task identifiers. Coverage is marginal, not conditional for every subgroup or time interval.

SplitCIFAR10, SplitCIFAR100, and CORe50 use frozen small-CNN feature representations. Low actionability may therefore reflect limitations of both the upstream representation and the continual learner; the present paper does not establish backbone sensitivity or invariance. Finally, the singleton gate is a conservative cross-benchmark baseline for autonomous single-label actionability, not a universal deployment policy or the maximum achievable action rate.

15 Reproducibility

Unlike v0.3.1, the v0.3.2 GitHub package contains the complete replay code and all seven compressed filtered traces. The original multi-hundred-megabyte evidence packages are not required for numerical reproduction and are omitted from Git. Exact filenames, checksums, and blank user-supplied Drive/archive location fields are provided for future provenance recovery.

16 Roadmap

The next stage is **MMALS-CAL v0.4: Online Regime Freshness and Change-Detection Harness**. It should compare fixed calibration, periodic recalibration, oracle change points, label-free CUSUM, delayed-label CUSUM, BOCPD, and adaptive conformal baselines. Evaluation must include false alarms, detection delay, coverage during transition, contamination, time to restored coverage, ACTION, wrong-action rate, and abstention cost.

17 Conclusion

MMALS-CAL adds a missing evidential layer to continual learning. It asks not only whether a system learned, but whether it recognized the situation, whether its uncertainty rule remains valid, whether the resulting set is compact, and whether action is justified. Calibration can control marginal coverage; it cannot manufacture competence, repair a wrong context, or turn a broad set into a useful decision. A trustworthy system must be able to say both "I can act" and "I do not yet have enough fresh evidence."

18 Annotated References: Abstracts, Engineer Translations, and Contributions to MMALS-CAL

Below, **"abstract" means a concise paraphrase of the work's contribution**, not a reproduction of the publisher's official abstract.

18.1 How the references fit together

Role in MMALS-CAL	References	Status in v0.3.2
Foundations of conformal prediction	Vovk et al.; Angelopoulos & Bates	Directly implemented
Coverage versus set compactness	Sadinle et al.	Direct conceptual support
Online adaptation under drift	Gibbs & Candès 2021/2022; Bhatnagar et al.	Roadmap, not yet implemented
Automatic regime-change detection	Page; Adams & MacKay	Roadmap component A
Fail-closed and constraint-oriented design	Thomas et al.	Architectural and verification philosophy

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18.2 1. Vovk, Gammerman, and Shafer, 2005

18.2.1 *Algorithmic Learning in a Random World*

Paraphrased abstract This foundational book develops conformal prediction as a way to attach statistically meaningful confidence information to machine-learning predictions. Instead of returning only a point prediction, a conformal predictor evaluates how unusual—or non-conforming—a candidate output is relative to previously observed examples. Under the classical randomness or exchangeability assumptions, the resulting prediction regions have controlled error rates without requiring the underlying predictive model to be probabilistically correct. The book connects these methods to algorithmic randomness, online prediction, confidence, and credibility. (link.springer.com)

Engineer translation You already have a classifier producing scores. Conformal prediction adds an external validation wrapper:

```
model scores
    ↓
nonconformity score
    ↓
calibration quantile
    ↓
prediction set with a measurable error rate
```

The important engineering idea is that **the classifier does not need to estimate perfect probabilities**. The calibration layer looks at how the scores behaved on held-out data and converts them into a prediction set.

For MMALS-CAL, this is the origin of:

$$s(x, y) = 1 - p(y \mid x)$$

and:

$$\Gamma(x) = \{y : s(x, y) \leq \widehat{q}\}.$$

What it brings to the MMALS-CAL paper This is the principal theoretical foundation for:

- the use of nonconformity scores;
- conformal calibration quantiles;
- prediction sets rather than unconditional single-label answers;
- marginal error control under exchangeability;
- separating the predictor from the uncertainty-verification layer.

It supports the claim that MMALS-CAL is **model-agnostic**: it can wrap MMALS outputs without modifying the MMALS learner.

Important limitation The classical validity result depends on calibration and deployment observations being exchangeable. It does not automatically remain valid when the operating regime changes.

That limitation is exactly why MMALS-CAL studies:

- stale calibration;
- global versus regime-specific calibration;
- context-selected calibrators;
- future online change detection.

18.3 2. Angelopoulos and Bates, 2021

18.3.1 *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification*

Paraphrased abstract This tutorial presents conformal prediction as a practical framework for converting the outputs of arbitrary pretrained models into prediction sets or intervals with explicit finite-sample coverage guarantees. It emphasizes that the method requires very few assumptions about the model or data distribution, while still requiring an appropriate exchangeability relationship between calibration and future observations. The paper provides practical examples, implementation guidance, evaluation methods, and extensions to structured prediction, distribution shift, abstention, and risk control. (arxiv.org)

Engineer translation This paper is the practical manual for turning conformal prediction into software.

The recipe is:

1. train or obtain a predictive model;
2. freeze it;
3. reserve a calibration dataset;
4. calculate one nonconformity score per calibration observation;
5. take a finite-sample corrected quantile;
6. use this quantile to form sets on new observations;
7. evaluate both coverage and set size.

It says, in effect:

> Do not redesign your neural network merely to obtain uncertainty estimates. Add a statistically controlled layer around its outputs.

What it brings to the MMALS-CAL paper It directly supports the practical implementation choices in v0.3.2:

- black-box wrapping of existing MMALS traces;
- held-out calibration data;
- finite-sample quantile correction;
- 90% target coverage;
- reporting set size alongside coverage;
- abstention and prediction-set evaluation.

It is probably the best citation for readers who want to understand **how MMALS-CAL works operationally** without first reading the deeper theoretical literature.

Important nuance "Distribution-free" does not mean "valid under every possible deployment shift."

It means that the method does not require a parametric distribution such as Gaussian noise. Classical validity still relies on exchangeability between calibration and deployment data.

18.4 3. Sadinle, Lei, and Wasserman, 2019

18.4.1 *Least Ambiguous Set-Valued Classifiers with Bounded Error Levels*

Paraphrased abstract The paper considers multiclass classifiers that return a set of plausible labels rather than a single label. It formulates the design problem as satisfying a user-specified coverage or error constraint while minimizing ambiguity, measured through the expected size of the returned label set. It derives optimal classifiers when the true class probabilities are known and proposes estimators with theoretical and finite-sample properties when those probabilities must be estimated. It also examines the possibility of empty prediction sets and practical ways to handle them. (arxiv.org)

Engineer translation Coverage alone is easy to achieve badly.

A system could always return:

```
{every possible class}
```

and obtain almost perfect coverage.

That would be statistically correct but operationally useless.

Sadinle et al. formalize the real engineering objective:

```
meet the error constraint
while
returning the smallest useful set
```

This produces two separate metrics:

- **validity:** is the true answer usually inside the set?
- **efficiency or ambiguity:** how many answers are inside the set?

What it brings to the MMALS-CAL paper This paper is the strongest support for the distinction between:

- **coverage**, and
- **compactness**.

It justifies why MMALS-CAL reports both:

$$\Pr(Y \in \Gamma(X))$$

and:

$$\mathbb{E}[|\Gamma(X)|].$$

It is directly relevant to the CORe50 result:

- approximately 90% coverage;
- average set size of about 28.7 labels out of 50;
- almost no singleton actions.

The reference helps us argue that **coverage without low ambiguity is not enough for actionability**.

Important distinction MMALS-CAL does not implement the exact optimization procedure proposed by Sadinle et al.

The paper supports our **compactness axis and engineering objective**, rather than serving as the exact algorithmic source of the LAC split-conformal method used in v0.3.2.

18.5 4. Gibbs and Candès, 2021

18.5.1 *Adaptive Conformal Inference Under Distribution Shift*

Paraphrased abstract The paper introduces adaptive conformal inference for online environments where the data-generating distribution changes over time. Instead of using one fixed calibration threshold indefinitely, the method continuously adjusts a parameter controlling prediction-set size according to recent coverage errors. The authors prove long-run coverage-frequency behavior without assuming a stationary or exchangeable sequence and demonstrate robustness under substantial real-world distribution shifts. (proceedings.neurips.cc)

Engineer translation Classical split conformal works like a fixed configuration:

```
calibrate once
→ obtain threshold
→ reuse threshold
```

Adaptive conformal inference adds a feedback loop:

```
produce prediction set
→ later receive true label
→ check whether coverage failed
→ widen or narrow future sets
```

It behaves like a controller:

- too many misses → become more conservative;
- too much excess coverage → potentially tighten the sets.

What it brings to the MMALS-CAL paper This reference supports the transition from:

offline evidence verification

to:

online adaptive calibration.

It is central to the proposed v0.4 roadmap.

It shows that the stale-calibrator problem observed experimentally in MMALS-CAL is not merely descriptive. There is an existing theoretical family of methods designed to update coverage control during changing conditions.

What it does not provide to v0.3.2 Adaptive conformal inference is not implemented in v0.3.2.

The current release:

- uses offline final-checkpoint traces;
- constructs fixed quantiles from reserved calibration subsets;
- compares fresh and stale quantiles;
- does not update quantiles sequentially.

Also, adaptive conformal inference is an adaptation mechanism, not necessarily a semantic detector that announces which named MMALS regime is active.

18.6 5. Gibbs and Candès, 2022

18.6.1 *Conformal Inference for Online Prediction with Arbitrary Distribution Shifts*

Paraphrased abstract This work extends adaptive conformal inference to respond more effectively to changing environments. It addresses the problem that excessive reliance on historical observations can make an online conformal method react too slowly. The proposed procedure dynamically tunes its adaptation rate and provides regret guarantees over local time intervals, allowing it to respond to distribution shifts of different sizes and speeds without prior knowledge of the shift dynamics. (arxiv.org)

Engineer translation The 2021 method needs an update speed:

- update too slowly → the system remains stale after a change;
- update too quickly → the threshold becomes noisy and unstable.

The 2022 method effectively manages several possible adaptation speeds and learns which one is useful.

Engineer analogy:

```
slow controller     → stable but sluggish
fast controller     → reactive but noisy
adaptive controller → chooses speed from recent behavior
```

What it brings to the MMALS-CAL paper It supports two important observations from our experiments:

1. **Calibration age matters.**
2. **Not all changes occur at the same speed.**

PermutedMNIST resembles a relatively abrupt functional shift, while RotatedMNIST suggests a more ordered or gradual geometric variation. A fixed adaptation speed may not be appropriate for both.

This paper is therefore relevant to:

- calibration freshness;
- local-window evaluation;
- adaptive recalibration speed;
- the future comparison between fixed, periodic, and adaptive calibration.

Important limitation This reference does not mean that MMALS-CAL v0.3.2 is already robust to arbitrary shifts.

It defines a candidate future method that should be compared against:

- fixed calibration;
- periodic recalibration;
- CUSUM-triggered recalibration;
- oracle change points.

18.7 6. Bhatnagar, Wang, Xiong, and Bai, 2023

18.7.1 *Improved Online Conformal Prediction via Strongly Adaptive Online Learning*

Paraphrased abstract The authors study online prediction sets when distributions may change arbitrarily. They argue that performance measured over the entire history can hide failures during shorter local intervals. They introduce strongly adaptive online conformal methods that control regret simultaneously over intervals of different lengths and obtain approximately valid coverage. Their experiments show improved local coverage and smaller prediction sets under distribution shift compared with earlier methods. (proceedings.mlr.press)

Engineer translation Imagine a system that is correct over one year on average but fails badly for two weeks after every regime change.

A global annual metric may still look acceptable.

Strongly adaptive methods ask:

> Did the system perform correctly on every meaningful local interval?

The proposed architecture uses multiple online experts, each active over different temporal intervals. This allows the system to adapt to:

- short abrupt shifts;
- long stable periods;
- medium-duration changes;

without choosing one fixed monitoring window in advance.

What it brings to the MMALS-CAL paper This reference strongly supports one of our main findings:

> Aggregate coverage can conceal local undercoverage.

MMALS-CAL already reports:

- global coverage;
- seed-level coverage;
- task-level coverage;
- worst seed-task coverage.

Bhatnagar et al. provide a theoretical online continuation of that idea: validity and efficiency should also be monitored over **local time windows**, not only over the complete run.

This is highly relevant to future metrics such as:

- worst-window coverage;
- local coverage error;
- recovery time after drift;
- window-dependent action rate.

Current versus future status MMALS-CAL v0.3.2 already adopts the **evaluation philosophy** of local auditing, but it does not implement SAOCP or strongly adaptive online learning.

18.8 7. Adams and MacKay, 2007

18.8.1 *Bayesian Online Changepoint Detection*

Paraphrased abstract This paper presents an online Bayesian method for detecting abrupt changes in the generative mechanism of sequential observations. The method maintains a probability distribution over the current run length—the time elapsed since the most recent changepoint. New observations update the posterior probability of each possible run length, allowing the system to represent both the likely position of a change and uncertainty about that position. (arxiv.org)

Engineer translation Rather than returning only:

```
change / no change
```

BOCPD returns something closer to:

```
70% probability that the current regime began 3 samples ago
20% probability that it began 8 samples ago
10% probability that no recent change occurred
```

Its internal state is the posterior distribution:

$$P(r_t \mid x_{1:t}),$$

where r_t is the current run length.

What it brings to the MMALS-CAL paper It provides the probabilistic design option for component A:

> automatic regime-change detection.

BOCPD would allow future MMALS-CAL versions to:

- identify suspected regime transitions;
- quantify uncertainty about the transition;
- estimate how long the current regime has existed;
- decide whether enough post-change observations have accumulated for calibration;
- trigger quarantine and buffer creation.

Important limitation BOCPD requires an explicit probabilistic model for observations and a hazard model for changepoints.

It is not distribution-free in the conformal-prediction sense.

For MMALS-CAL, a design question remains:

> What signal should BOCPD observe?

Candidates include:

- prediction entropy;
- context entropy;
- route drift;
- embedding drift;
- delayed-label nonconformity;
- empirical miscoverage.

18.9 8. Page, 1954

18.9.1 *Continuous Inspection Schemes*

Paraphrased abstract Page studies sequential monitoring procedures designed to detect when an observed process has moved away from its expected operating condition. The central idea underlying CUSUM is to accumulate small deviations over time so that a persistent shift becomes detectable even when each individual observation is not sufficiently unusual to trigger an alarm. The method balances detection delay against false alarms through its reference value and alarm threshold. (jstor.org)

Engineer translation A single strange observation may be noise.

Ten slightly strange observations in the same direction may indicate a real regime change.

CUSUM keeps a running statistic such as:

$$S_t = \max(0, S_{t-1} + z_t - k),$$

and raises an alarm when:

$$S_t > h.$$

Where:

- z_t is the monitored deviation;
- k ignores small expected fluctuations;
- h controls alarm sensitivity.

What it brings to the MMALS-CAL paper CUSUM is the simplest interpretable candidate for component A.

It is attractive because it is:

- incremental;
- computationally cheap;
- easy to audit;
- suitable for real-time systems;
- explicit about the false-alarm versus detection-delay trade-off.

For MMALS-CAL, it could monitor separate channels:

```
CUSUM on class uncertainty
CUSUM on context uncertainty
CUSUM on route drift
CUSUM on embedding drift
CUSUM on delayed-label nonconformity
```

Important limitation CUSUM detects sustained change in a chosen signal. It does not explain the semantic meaning of the new regime, and it is only as good as:

- the monitored statistic;
- the reference distribution;
- the selected threshold;
- the assumed shift direction.

It should therefore not be confused with MMALS context inference itself.

18.10 9. Thomas et al., 2019

18.10.1 *Preventing Undesirable Behavior of Intelligent Machines*

Paraphrased abstract The paper proposes a framework for designing machine-learning algorithms around explicit user-defined constraints on undesirable behavior. Rather than expecting users to indirectly control risk through penalties or hyperparameters, the framework asks them to specify the behavior that must be avoided and the acceptable probability of violation. The resulting Seldonian approach may refuse to return a solution when the available evidence is insufficient to support the required constraint. The authors demonstrate the approach on examples involving harmful or unfair behavior. (science.org)

Engineer translation Normal ML design asks:

> Which model maximizes performance?

The Seldonian design question is:

> Which model maximizes performance while giving us enough statistical evidence that an undesirable condition will not be violated?

And when evidence is insufficient:

```
do not return a supposedly safe solution
```

This is the same general fail-closed philosophy as:

```
insufficient evidence
→ NO_DECISION
```

What it brings to the MMALS-CAL paper This reference supports the **system-design philosophy**, especially:

- explicit undesirable outcomes;
- evidence-based constraints;
- separating candidate construction from independent verification;
- refusing to act when the constraint cannot be supported;
- moving responsibility from the end user to the algorithm and protocol designer.

It also connects strongly to the MMALS Verification Stack:

```
candidate result
    ↓
protocol verification
    ↓
statistical safety/evidence test
    ↓
release, ACTION, or refusal
```

Critical distinction MMALS-CAL is **not currently a Seldonian algorithm**.

The guarantee types differ:

- conformal prediction controls marginal coverage of prediction sets;
- Seldonian methods seek high-probability satisfaction of explicitly specified behavioral constraints.

Thomas et al. support our **fail-closed architecture and claim discipline**, but they should not be cited as proof that MMALS-CAL provides broad machine-safety guarantees.

18.11 What the references collectively bring to MMALS-CAL

18.11.1 1. They establish the statistical foundation

Vovk et al. and Angelopoulos–Bates justify the transformation:

```
model output
→ calibrated prediction set
→ measurable marginal coverage
```

Without these references, MMALS-CAL could look like an ad hoc confidence-thresholding mechanism.

With them, it is situated within conformal prediction.

18.11.2 2. They justify the five-axis decomposition

Sadinle et al. show why coverage must be separated from ambiguity.

That directly supports:

```
Competence
Coverage
Compactness
Context
ACTION
```

A system can satisfy coverage while being non-actionable because its sets are too large.

18.11.3 3. They explain why v0.3.2 is deliberately offline

Classical split conformal gives the cleanest experimental foundation under controlled calibration/deployment separation.

The online papers demonstrate that the next step is possible, but also that it introduces new questions:

- adaptation rate;
- local-window validity;
- delayed labels;
- temporal dependence;
- calibration recovery;
- shift speed.

Thus, v0.3.2 validates components B, C, and a simplified D before adding component A.

18.11.4 4. They provide two different ways to handle changing regimes

Explicit change detection

- Page: lightweight cumulative alarm;
- Adams–MacKay: probabilistic run-length inference.

Continuous calibration adaptation

- Gibbs–Candès: feedback adaptation;
- FACI-style adaptation to shift speed;
- Bhatnagar et al.: interval-local strongly adaptive methods.

These are complementary rather than identical:

```
change detection asks:
"Did the regime change?"

adaptive conformal asks:
"How should the prediction-set threshold evolve?"
```

A complete runtime MMALS-CAL may eventually need both.

18.11.5 5. They justify fail-closed behavior

Thomas et al. support the principle that an intelligent system should be allowed to return:

```
insufficient evidence
```

rather than being forced to provide an answer.
This gives stronger intellectual grounding to:

```
ACTION / NO_DECISION
```

and to MMALS-CAL's prohibited-claims discipline.

18.12 Recommended citation placement in the paper

Paper section	References
Introduction to conformal prediction	Vovk et al.; Angelopoulos & Bates
Calibration quantile and exchangeability	Vovk et al.; Angelopoulos & Bates
Coverage versus compactness	Sadinle et al.
Stale evidence and distribution shift	Gibbs & Candès 2021
Future adaptive calibration	Gibbs & Candès 2021, 2022; Bhatnagar et al.
Regime-change detector roadmap	Page; Adams & MacKay
ACTION/NO_DECISION and fail-closed design	Thomas et al.
Limitations	All online references, explicitly marked as not implemented

18.12.1 Suggested literature-positioning paragraph

MMALS-CAL builds on three complementary research traditions. Classical conformal prediction provides the statistical foundation for converting black-box model scores into prediction sets with finite-sample marginal coverage under exchangeability. Set-valued classification further establishes that coverage alone is insufficient and that ambiguity, measured through prediction-set size, must be controlled separately. Online conformal inference and changepoint-detection methods provide the foundations for the future transition from offline replay to runtime calibration freshness, while Seldonian algorithm design motivates the fail-closed principle that a system should refuse to act when the available evidence cannot support its declared constraint.

The current v0.3.2 release implements offline no-leakage calibration, regime-aware conformal quantiles, compactness auditing, and a conservative singleton ACTION/NO_DECISION baseline. It does not yet implement CUSUM, Bayesian online changepoint detection, adaptive conformal inference, FACI, or strongly adaptive online conformal prediction. These references therefore serve both as foundations for the implemented verification layer and as explicit methodological anchors for the next online stage.

The most important message is that the bibliography is not one undifferentiated list. It describes the evolution of the project:

offline conformal validity
→ compactness and actionability
→ regime-change detection
→ online adaptive calibration
→ fail-closed operational control.

References

[1] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

[2] A. Angelopoulos and S. Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511, 2021.

[3] M. Sadinle, J. Lei, and L. Wasserman. Least ambiguous set-valued classifiers with bounded error levels. JASA, 2019.

[4] I. Gibbs and E. Candes. Adaptive conformal inference under distribution shift. NeurIPS, 2021.

[5] I. Gibbs and E. Candes. Conformal inference for online prediction with arbitrary distribution shifts. arXiv:2208.08401, 2022.

[6] A. Bhatnagar et al. Improved online conformal prediction via strongly adaptive online learning. ICML, 2023.

[7] R. Adams and D. MacKay. Bayesian online changepoint detection. arXiv:0710.3742, 2007.

[8] E. Page. Continuous inspection schemes. Biometrika, 1954.

[9] P. Thomas et al. Preventing undesirable behavior of intelligent machines. Science, 2019.